

T-JOHN INSTITUTE OF TECHNOLOGY

SkillIntel

AI-Powered Career Intelligence Platform

Smarter skill analysis, career guidance and placement readiness for every student

PROJECT TEAM

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Review Date: 13 March 2026

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Introduction

The placement-readiness problem

A

Competitive Job Market

Students struggle to identify the exact skills industry employers expect today.

B

Marks-Only Mindset

Most students focus only on academic marks and ignore real industry-readiness signals.

C

Low Placement Readiness

Without skill awareness, demand insight or career strategy, students stay underprepared.

UNAWARE OF

Skill gaps

Industry demand

Career preparation strategies

Result: low placement readiness

Proposed Solution

SkillIntel — an AI career intelligence platform

SkillIntel helps students prepare effectively for placements through AI-driven skill analysis, career recommendations and interview readiness — all in one platform.

01 Analyze Skills

Map current student competencies against industry roles

02 Identify Gaps

Highlight missing skills that hold back placement

03 Recommend Careers

Match students to suitable AI-curated career paths

04 Evaluate Resumes

ATS-style scoring and improvement guidance

05 Coding Practice

Curated problems and quizzes to sharpen skills

06 Interview Prep

Common interview questions and mock practice

Literature Survey

Existing systems for career preparation

LinkedIn Learning

Provides courses but lacks personalized skill analysis.

Course Platform

HackerRank / LeetCode

Strong coding practice only — no career guidance layer.

Coding Practice

Resume Analyzers

Provide ATS scoring but no broader career direction.

ATS Tools

College Placement Portals

Track placements but lack analytics and predictions.

Tracking

Insight: each tool solves one piece — none integrate skills, careers, resumes and analytics in one place.

Limitations of Existing Systems

Why a unified platform is needed

1

Tools are not integrated

Students switch between many disjoint platforms

2

No skill gap analysis

No insight into what students are actually missing

3

No career recommendation

Lack of guidance toward suitable job roles

4

No readiness prediction

Cannot forecast placement readiness in advance

5

Limited student analytics

Faculty lack visibility into individual progress

6

Need for SkillIntel

Unifies analysis, prediction and guidance in one platform

Project Gaps

Major problems SkillIntel addresses

1

Students don't know which skills they lack

2

Lack of centralized career preparation systems

3

Limited placement readiness tracking

4

Faculty cannot easily monitor student progress

SKILLINTEL ANSWER

AI-Driven Analytics & Career Intelligence

A single platform that surfaces gaps, predicts readiness and guides students, faculty and placement teams together.

These gaps make career preparation reactive — SkillIntel makes it predictive and personalized.

Objective

Main goal and specific objectives

MAIN OBJECTIVE

To develop an AI-powered student career intelligence platform that analyzes academic performance, skills and learning behavior to improve placement readiness.

01

Identify Skill Gaps

Use data analysis to surface what each student is missing

02

Predict Job Readiness

Score readiness from academics, skills and projects

03

Personalized Learning

Recommend resources and career paths per student

04

Faculty Monitoring

Enable progress and performance analytics for staff

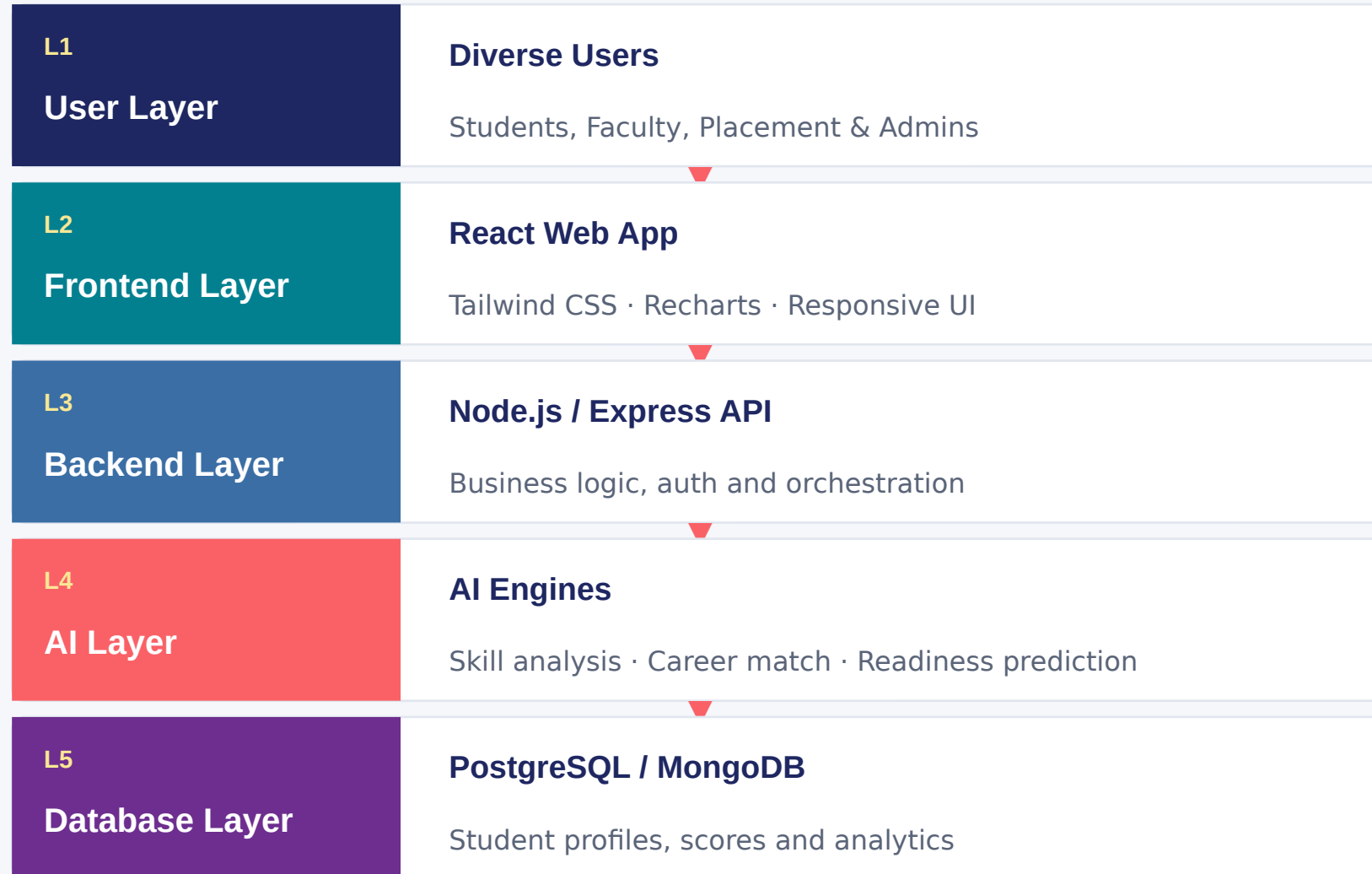
05

Placement Support

Help teams identify job-ready candidates fast

Overall Architecture

SkillIntel system layers — end-to-end flow



OUTPUT LAYER

Dashboards & Reports

Student Dashboard

Faculty Dashboard

Placement Dashboard

Skill Gap Reports

Career Recommendations

Technologies Used

Stack, libraries and developer tools

FRONTEND

React.js

Component-based UI library

Tailwind CSS

Utility-first styling

BACKEND

Node.js

JavaScript runtime

Express.js

REST API framework

DATABASE

PostgreSQL

Relational analytics store

MongoDB

Flexible document store

LIBRARIES & TOOLS

Recharts

Data visualization

pdfjs-dist

Resume parsing

VS Code · GitHub · Vercel

Dev & deployment

Project Timeline

12-week development plan



Expected Output

What SkillIntel will deliver

01

Student Dashboard

Academic performance, skills and job readiness score

02

Skill Gap Analysis

Missing skills mapped to target career roles

03

AI Career Recommendations

Suitable job roles based on profile

04

Resume ATS Scoring

Improvement suggestions for each section

05

Coding Practice & Quizzes

Evaluate technical skills with results

06

Interview Preparation

Common questions and mock practice

07

Readiness Prediction

Forecast placement readiness from data

08

Faculty & Placement Dashboards

Performance analytics for teams

09

Learning Resources

Curated content for skill improvement

Conclusion

SkillIntel impact across roles

After implementation, SkillIntel will deliver role-specific intelligence that turns raw academic data into actionable career insights for students, faculty and placement teams.

Student Dashboard

Skill analysis

Job readiness score

Career recommendations

Faculty Dashboard

Student analytics

Skill gap reports

Progress monitoring

Placement Dashboard

Readiness prediction

Performance analytics

Job-ready candidate list

Thank You



Questions & Discussion